

TREE-01

Q3)

Question:

DS	Session	Tree 1	Question Information	Level 1	Challenge 63
Question description					
You are given an $n \times n$ grid representing the map of a forest. Each square is either empty or contains a tree. The upper-left square has coordinates $(1,1)$, and the lower-right square has coordinates (n,n) .					
Your task is to process q queries of the form: how many trees are inside a given rectangle in the forest?					
Constraints					
$1 \leq n \leq 1000$ $1 \leq q \leq 2 \cdot 10^5$ $1 \leq y1 \leq y2 \leq n$ $1 \leq x1 \leq x2 \leq n$					
Input					
The first input line has two integers n and q : the size of the forest and the number of queries.					
Then, there are n lines describing the forest. Each line has nn characters: $.$ is an empty square and $*$ is a tree.					
Finally, there are q lines describing the queries. Each line has four integers $y1, x1, y2, x2$ corresponding to the corners of a rectangle.					
Output					
Print the number of trees inside each rectangle.					

PROGRAM:

```
1  #include <stdio.h>
2
3  int a[1005][1005];
4
5  int main()
6  {
7      int n, q, i, j;
8      char s[1005];
9
10     scanf("%d %d", &n, &q);
11
12     for (i = 1; i <= n; i++)
13     {
14         scanf("%s", s);
15
16         for (j = 1; j <= n; j++)
17         {
18             a[i][j] = (s[j - 1] == '*');
19             a[i][j] = a[i][j] + a[i - 1][j] + a[i][j - 1] - a[i - 1][j - 1];
20         }
21     }
22
23     while (q--)
24     {
25         int y1, x1, y2, x2;
26
27         scanf("%d %d %d %d", &y1, &x1, &y2, &x2);
28
29         printf("%d\n", a[y2][x2] - a[y1 - 1][x2]
30             - a[y2][x1 - 1] + a[y1 - 1][x1 - 1]);
31     }
32
33     return 0;
34 }
```

OUTPUT:

TEST CASE 1:

```
4 3
.*.
*.*
*.*
*.*
****
2 2 3 4
3
3 1 3 1
1
1 1 2 2
2
```

TEST CASE 2:

```
4 3
.*.*
*.*
*.*
*.*
****
1 2 3 4
6
3 2 3 4
2
1 4 2 3
0
```